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Additive Manufacturing (3D printing)

microfluidic device is fabricated layer by layer.

CADworks Profluidics 285D DLP printer

1. Create a microfluidic device in CAD software (Solidworks or Fusion). Include channel network, inlet and outlet ports, reservoirs, etc.
2. Export the model as an STL file
3. Connect to the PROFluidics 285D, import the STL, arrange on build platform, select appropriate material.
4. Ensure build platform and resin vat are clean and that resin is available. Wear PPE
5. Start the job, and once device is completed, wash away uncured resin using solvent and use fine syringe to flush the channels.
6. UV cure the ~~device (resin)~~ microfluidic

If you are using printer to print a mold & not device
* using a mold-specific resin (like Master mold). Print the mold, cast PDMS against it, cure PDMS, remove from mold. This produces seperate, flexible microfluidic device.

3D-printer → mold → PDMS → microfluidic device

vs.

3D-printer → microfluidic device